PEM 2026 Software Engineering Trial Examination Answers

Section I

Question	Answer Key
1	D
2	B
3	B
4	C
5	B
6	C
7	C
8	C
9	B
10	A
11	B
12	D
13	C
14	D

15.

Feature	FTP	HTTP
Used to transfer web pages between a browser and a server	☐	√
Commonly used for uploading and downloading files	√	☐
Uses port 80 by default	☐	√
Transfers data in plain text by default	√	☐

16. A, C, E

SECTION II

Q17.

Marking criteria	Marks
Clearly outlines a valid difference between AI and ML using correct terminology	2
Identifies a difference but lacks clarity or accuracy	1

Sample answer : Artificial Intelligence is a broad field focused on creating systems that perform tasks requiring human intelligence, while machine learning is a subset of AI that allows systems to learn patterns from data and improve performance without being explicitly programmed.

Q18.

Marking criteria	Marks
Identifies and explains a valid benefit of data protection	2
Identifies a benefit with limited explanation	1

Sample answer: Data protection prevents unauthorised access to sensitive information, reducing the risk of data breaches and protecting user privacy.

Q19:

Marking criteria	Marks
Explains a communication issue and clearly discusses its impact on a project	3
Identifies a relevant issue with some explanation	2
Identifies an issue with minimal explanation	1

Sample answer: Poor communication can cause misunderstandings about project requirements. If stakeholders are not regularly consulted, the final product may fail to meet user expectations, leading to rework and increased costs.

Q20

Marking criteria	Marks
Clearly justifies PWA use with three relevant benefits linked to the scenario	3
Provides two relevant benefits linked to the scenario	2
Provides one relevant benefit	1

Sample answer: A PWA is suitable because it works offline, reducing issues caused by unreliable internet. It requires less storage than native apps and works across multiple platforms, making it cost-effective for a business with limited resources.

Q21

Marking criteria	Marks
Accurately compares both SAST and DAST, clearly identifying differences	3
Describes both techniques with limited comparison	2
Describes one technique or provides minimal comparison	1

Sample answer: SAST tests the source code without running the application and identifies vulnerabilities early in development. DAST tests an application while running and identifies security vulnerabilities at runtime.

Q22

Marking criteria	Marks
Clearly explains an ethical issue and its implications	3
Identifies an ethical issue with some explanation	2
Identifies an ethical issue with minimal explanation	1

Sample Answer: Bias in training data can result in unfair decisions by automated systems. If historical data contains discrimination, the model may reinforce inequality, reducing trust and fairness.

Q23

Marking criteria	Marks
Clearly describes how ML supports automation and explains its benefits	3
Identifies ML use with some explanation	2
Identifies ML use with minimal explanation	1

Sample Answer: (DevOps example)

ML supports DevOps by predicting system failures through monitoring data. This allows issues to be fixed proactively, improving reliability and reducing downtime.

Q24

Marking criteria	Marks
Explains multiple key features of Privacy by Design relevant to software security	3
Explains one or two features with limited detail	2
Identifies Privacy by Design with minimal explanation	1

Sample Answer

Privacy by Design ensures data protection is built into systems from the start. Key features include data minimisation, secure defaults, and encryption, reducing privacy risks.

Q25

Marking criteria	Marks
Accurately compares both learning types and provides relevant examples	3
Compares learning types with limited explanation or examples	2
Identifies a difference without clear examples	1

Sample Answer: Supervised learning uses labelled data, such as spam detection. Unsupervised learning finds patterns in unlabelled data, such as customer segmentation.

Q26

Marking criteria	Marks
Clearly explains the purpose of session management with reference to security and user state	3
Explains session management with limited detail	2
Minimal understanding demonstrated	1

Sample Answer: Session management maintains user state across requests, improves security by authenticating users, and prevents unauthorised access.

Q27

Marking criteria	Marks
Correctly identifies Agile (or appropriate iterative approach)	1

Sample Answer:

(i)Agile development

(ii)

Marking criteria	Marks
Justifies approach using clear references to scenario requirements	3
Provides general justification with limited scenario link	2
Minimal justification provided	1

Sample Answer:Agile allows features to be delivered early and refined based on feedback, which suits a project with evolving requirements and user needs.

Q28

Marking criteria	Marks
Accurately describes sandboxing in secure software architecture	3
Describes sandboxing with limited accuracy	2
Minimal description	1

Sample Answer: Sandboxing isolates code execution in a controlled environment to prevent access to critical system resources. It is used to safely run untrusted code, such as user-uploaded scripts, reducing security risks.

Q29

Marking criteria	Marks
Clearly explains testing’s role in quality, reliability, and defect detection	3
Explains testing with limited detail	2
Minimal understanding demonstrated	1

Sample Answer: Testing identifies defects early, ensures requirements are met, and improves reliability by preventing failures in production.

Q 30

User_id *int* *Customer identifier number*

Customer_fullname

Product_name *Str*

Prices *int*

Order_status *boolean*

Q31

Marking criteria	Marks
Clearly explains overfitting and provides a relevant example	3
Explains overfitting but example is weak or unclear	2
Minimal explanation or example	1

Sample Answer: Overfitting occurs when a model learns training data too well and performs poorly on new data, such as a fraud model trained on outdated patterns.

Q32

a.

Marking criteria	Marks
Correct Python statement producing correct output	2
Minor syntax or logic error	1

Sample answer: print(sum(prices))

b.

Marking criteria	Marks
Correct function definition and parameters	1
Correct total calculation	2
Discount condition correctly implemented	1
Output formatted correctly	1
Code readable and meets all requirements	1

Sample answer:

```
def process_order(customer_name, prices):
total = sum(prices)
if total > 30:
    total *= 0.9
total = round(total, 2)
print(f"Customer: {customer_name}")
print(f"Final cost: ${total}")
return total
```

Q33

Marking criteria	Marks
Sustained, well-reasoned discussion addressing benefits and challenges, linked to scenario	5–6
Clear discussion with some analysis	3–4
Basic description with limited evaluation	1–2

Sample answer: Machine learning may displace routine roles but create new opportunities in system oversight and data analysis. While efficiency increases, reskilling is essential to reduce inequality.

Software Engineering 2026 Trial Exam Mapping Grid

Question(s)	Syllabus Outcome(s)	Focus Area	Band Alignment
1	**EC-12-02**, **EC-12-08**	Programming for the Web	Band 3–4
2	**EC-12-02**, **EC-12-08**	Project Management	Band 3–4
3	**EC-12-06**	Software Automation	Band 4
4	**EC-12-01**, **EC-12-02**	Programming for the Web	Band 3–4
5	**EC-12-02**	Programming for the Web	Band 3
6	**EC-12-10**, **EC-12-09**	Software Engineering Project	Band 4
7	**EC-12-10**, **EC-12-09**	Software Engineering Project	Band 4
8	**EC-12-08**	System development tools	Band 3–4
9	**EC-12-09**	Project Management	Band 4
10	**EC-12-02**	Data use and manipulation	Band 4
11	**EC-12-06**	Intelligent systems	Band 5
12	**EC-12-01**	Project Management	Band 4
13	**EC-12-05**	Programming for the Web	Band 4
14	**EC-12-02**	Programming for the Web	Band 3
15	**EC-12-02**	Programming for the Web	Band 4–5
16	**EC-12-06**	Software Automation	
17	**EC-12-06**	Software Automation	Band 4
18	**EC-12-03**	Software Secure Architecture	Band 4
19	**EC-12-09**	Project management	Band 4
20	**EC-12-08**, **EC-12-01**	Programming for the Web	Band 5
21	**EC-12-03**	Software Secure Architecture	Band 5
22	**EC-12-07**	Software Automation	Band 5–6
23	**EC-12-06**, **EC-12-01**	Software Automation	Band 5–6
24	**EC-12-03**, **EC-12-07**	Software Secure Architecture	Band 5
25	**EC-12-06**	Software Automation	Band 4–5
26	**EC-12-03**, **EC-12-03**	Software Secure Architecture	Band 4
27	**EC-12-08**, **EC-12-09**	Software Engineering Project	Band 5
28	**EC-12-03**	Secure system architecture	Band 5
29	**EC-12-10**, **EC-12-09**	Software Engineering Project	Band 4–5
30	**EC-12-02**, **EC-12-08**	Software Engineering Project	Band 4
31	**EC-12-06**	Software Automation	Band 5
32	**EC-12-08**	Software Engineering Project	Band 5–6
33	**EC-12-07**, **EC-12-06**	Software Automation	Band 6